

CS1231S: Discrete Structures
Tutorial #5: Relations & Partial Orders
(Week 7: 28 September – 2 October 2026)

I. Discussion Questions

- D1. Let R be a binary relation on a non-empty set A . If $R = \emptyset$, then is R reflexive? Symmetric? Transitive?
- D2. Suppose a binary relation R on a non-empty set A is reflexive, transitive, symmetric and antisymmetric. What can you conclude about R ? Explain.
- D3. Asymmetry is defined in question 6 as follows: A binary relation R on a set A is **asymmetric** iff
$$\forall x, y \in A (x R y \Rightarrow y \not R x).$$

Are there binary relations that are both symmetric and asymmetric?
- D4. Let R be a relation on A . Prove that the following is an alternative definition of **antisymmetry**:
$$\forall x, y \in A ((x \neq y) \Rightarrow ((x, y) \in R \Rightarrow (y, x) \notin R)).$$

II. Tutorial Questions

1. Let S be the set of all strings over the alphabet $\mathcal{A} = \{s, u\}$, i.e. an element of S is a sequence of characters, each of which is either s or u . Examples of elements of S are: ε (the empty string), $s, u, sus, usssuu$, and $susussssu$.

Define a relation R on S by the following: $\forall a, b \in S (a R b \Leftrightarrow \text{len}(a) \leq \text{len}(b))$
where $\text{len}(x)$ denotes the length of x , i.e. the number of characters in x .

Is R a partial order? Prove or disprove it.

2. [AY2022/23 Semester 1 Midterm Test]
Let $A = \{2, 3, 5, 7, 21, 30, 84, 99\}$ and let $\preceq$ be a partial order on the set A defined by the “divides” relation, that is, $x \preceq y \Leftrightarrow x|y$. Which of the following statements are true?
- (i) The partial order has a linearization $\preceq^*$ such that $21 \preceq^* 7$.
 - (ii) The partial order has a linearization $\preceq^*$ such that $3 \preceq^* 2$.
 - (iii) The partial order has a linearization $\preceq^*$ such that $21 \preceq^* 5 \preceq^* 84$.
 - (iv) The partial order has a linearization $\preceq^*$ such that $99 \preceq^* 84 \preceq^* 30$.

3. [AY2023/24 semester 1 Midterm Test]
Let $A = \{11, 12, 13, 14, 15, 16\}$. For each $x \in A$, define $F_x = \{k \in \mathbb{Z}^+ : k|x\}$, where $|$ is the “divides” relation. Define also a partial order $\preceq$ on A by setting for all $x, z \in A$:

$$x \preceq z \Leftrightarrow (F_x = F_z) \vee (|F_x| < |F_z|).$$

What are the minimal, smallest, maximal, and largest elements of A with respect to $\preceq$?

4. Given the partial order $\leq$ on A in question 3 above, one of the linearizations $\leq^*$ of $\leq$ is shown below:

$$11 \leq^* 13 \leq^* 14 \leq^* 15 \leq^* 16 \leq^* 12$$

Write out all the other possible linearizations $\leq^*$.

5. Let $B = \{0,1\}$ and define the binary relation R on $B \times B$ as follows:

$$\forall (a,b), (c,d) \in B \times B \left((a,b) R (c,d) \Leftrightarrow (a \leq c) \wedge (b \leq d) \right).$$

- (a) Prove that R is a partial order.
- (b) Draw the Hasse diagram for R .
- (c) Find the maximal, largest, minimal and smallest elements.
- (d) Is $(B \times B, R)$ well-ordered?

6. Let R be a binary relation on a non-empty set A . Let $x, y \in A$. Define a relation S on A by

$$x S y \Leftrightarrow (x = y) \vee (x R y) \text{ for all } x, y \in A.$$

Show that:

- (a) S is reflexive;
- (b) $R \subseteq S$; and
- (c) if S' is another reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$.

What is this relation S called? (Hint: Refer to Transitive Closure in Lecture 6).

7. Let R be a binary relation on a set A .

We have defined antisymmetry in class: R is **antisymmetric** iff

$$\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y).$$

We define asymmetry here. R is **asymmetric** iff

$$\forall x, y \in A (x R y \Rightarrow y \not R x).$$

- (a) Find a binary relation on A that is both asymmetric and antisymmetric.
- (b) Find a binary relation on A that is not asymmetric but antisymmetric.
- (c) Find a binary relation on A that is asymmetric but not antisymmetric.
- (d) Find a binary relation on A that is neither asymmetric nor antisymmetric.

8. **Definitions.** Consider a partial order $\leq$ on a set A and let $a, b \in A$.

- We say a, b are **comparable** iff $a \leq b$ or $b \leq a$.
- We say a, b are **compatible** iff there exists $c \in A$ such that $a \leq c$ and $b \leq c$.

Consider the “divides” relation on $A = \{1, 2, 4, 5, 10, 15, 20\}$. List out the pairs of distinct elements in A that are (a) comparable; (b) compatible. Use the notation $\{x, y\}$ to represent the pair of elements x and y .

Are two comparable elements always compatible? Are two compatible elements comparable?

9. [AY2022/23 Semester 2 Mid-term Test]

Let $\leq$ be a partial order on a non-empty set A . A subset C of A is called a **chain** if and only if every pair of elements in C is comparable, that is, $\forall a, b \in C (a \leq b \vee b \leq a)$. A **maximal chain** is a chain M such that $t \notin M \Rightarrow M \cup \{t\}$ is not a chain. The length of a chain is one less than the number of elements in it.

- (a) Let $A = \{a, b, c, d\}$ and $(\mathcal{P}(A), \subseteq)$ be a poset on $\mathcal{P}(A)$, where $\mathcal{P}(A)$ denotes the power set of A . Write out two maximal chains in $(\mathcal{P}(A), \subseteq)$.
- (b) Let $B = \{2, 3, 5, 6, 7, 11, 12, 35, 385\}$ and $(B, |)$ be a poset on B , where $|$ denotes the divides relation. Draw the Hasse diagram and write out two maximal chains of different lengths in $(B, |)$.